

ENTITY ANALYSIS

CHAPTER – 2

2 INTRODUCTION

This section identifies the entities, attributes, primary keys, and relationships required for the HUFT ECommerce Order Management System. These components help define the structure of the database and provide the foundation for designing an efficient and organized Database Management System (DBMS).

2.1 ENTITY IDENTIFICATION

The following entities have been identified for the Heads Up For Tails (HUFT) E-Commerce Order Management System. These entities represent the core components of the database and are essential for managing customer information, products, orders, payments, and deliveries.

2.2 ENTITY ATTRIBUTES

Entity	Description
Customer	Stores customer details.
Product	Stores product information.
Category	Stores product categories.
Order	Stores customer order details.
Order Item	Stores products included in each order.
Payment	Stores payment information.
Shipment	Stores delivery details.
Admin	Stores administrator details.

2.2.1 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Contact number

Attribute	Description
Address	Delivery address

2.2.2 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Price	Product price
Stock	Available stock
Category_ID	Category ID

2.2.3 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name

2.2.4 ORDER TABLE

Attribute	Description
Order_ID	Unique product ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

2.2.5 ORDER ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2.2.6 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Status	Payment status

2.2.7 SHIPMENT TABLE

Attribute	Description
Shipment_ID	Unique shipment ID
Order_ID	Order ID
Tracking_Number	Tracking number
Delivery_Status	Delivery status

2.2.8 ADMIN TABLE

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address

Attribute	Description
Password	Login password

2.3 PRIMARY KEYS

Entity	Primary Key
Customer	Customer_ID
Product	Product_ID
Category	Category_ID
Order	Order_ID
Order Item	Order_Item_ID
Payment	Payment_ID
Shipment	Shipment_ID
Admin	Admin_ID

2.4 FOREIGN KEYS

Table	Foreign Key	References
Product	Category_ID	Category
Order	Customer_ID	Customer
Order Item	Order_ID	Order
Order Item	Product_ID	Product
Payment	Order_ID	Order
Shipment	Order_ID	Order

2.5 RELATIONSHIP

Parent Entity	Relationship	Child Entity
Customer	One-to-Many	Order
Category	One-to-Many	Product
Order	One-to-Many	Order Item
Product	One-to-Many	Order Item
Order	One-to-One	Payment
Order	One-to-One	Shipment

2.5.1 RELATIONSHIP EXPLANATION

Relationship	Cardinality	Explanation
Customer → Order	One-to-Many (1:M)	One customer can place multiple orders.
Category → Product	One-to-Many (1:M)	One category can contain multiple products.
Order → Order Item	One-to-Many (1:M)	One order can include multiple products.
Product → Order Item	One-to-Many (1:M)	One product can appear in multiple orders.
Order → Payment	One-to-One (1:1)	Each order has one payment record.
Order → Shipment	One-to-One (1:1)	Each order has one shipment record.

2.6 CONCLUSION

The identified entities, attributes, primary keys, foreign keys, and relationships provide the foundation for designing the E-Commerce Order Management Database System for Heads Up For Tails (HUFT). These components will be used to create the Entity Relationship (ER) Diagram and the relational database design in the next phase of the project.